Arista Access Node WFD-ACC-14 Whitefield POP Hardware/Environmental Fault Recovery SOP

Node: WFD-ACC-14 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_3211

Description

This Standard Operating Procedure outlines the recovery procedures for a major hardware/environmental fault on Arista Access Node WFD-ACC-14 located at the Whitefield POP. The failure is caused by an ASIC memory parity error or a line card crash, resulting in a loss of power supply redundancy, triggering Alarm Code ALARM_3211. This scenario is a critical failure domain due to the potential for widespread network outages and data loss. The underlying cause is attributed to the hardware design of the Arista 7280SR-48YC6 switch, which utilizes a combination of ASICs to handle forwarding, routing, and packet processing. The loss of memory parity or a line card crash can lead to a cascade of failures, impacting the control and data planes of the network. In this scenario, the access role is impacted, and the failure is classified as major due to the potential for severe network disruptions.

Pre-requisite Checks

- pre-requisite: Verify that the Whitefield POP is accessible and that remote hands are dispatched to the site.

- pre-requisite: Check the status of the redundant links and verify that they are operational.

- pre-requisite: Obtain confirmation from the NOC team that the alarm has been escalated and that a maintenance window has been scheduled.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal memory, Expected Output: System internal memory information, including memory parity errors, Description: This command will display the system internal memory information, including any memory parity errors that may be present.

Command: show system internal linecard, Expected Output: Line card status, including any crashes or errors, Description: This command will display the line card status, including any crashes or errors that may be present.
```

```
Command: show power redundancy, Expected Output: Power supply redundancy status, Description: This command will display the power supply redundancy status, including any losses or errors.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: hw-module replace , Description: Replace the faulty ASIC or line card with a new one, following the standard replacement procedures.

Step 2: Step: Reboot the switch, Command: reload, Description: Reboot the switch to ensure that all changes take effect and that the system is stable.

Step 3: Step: Verify power supply redundancy, Command: show power redundancy, Description: Verify that the power supply redundancy is restored and that there are no errors or losses.

Post-Check Verification

```
Command: show system internal memory, Expected Output: System internal memory information, including memory parity errors, Description: Verify that the system internal memory information is correct and that there are no memory parity errors.

Command: show system internal linecard, Expected Output: Line card status, including any crashes or errors, Description: Verify that the line card status is correct and that there are no crashes or errors.

Command: show power redundancy, Expected Output: Power supply redundancy status, Description: Verify that the power supply redundancy status is correct and that there are no errors or losses.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy running-config startup-config, Description: Restore the original configuration to ensure that the system is returned to its previous state.

- Step: Reboot the switch, Command: reload, Description: Reboot the switch to ensure that all changes take effect and that the system is stable.

- Step: Verify system integrity, Command: show system internal memory, Description: Verify that the system internal memory information is correct and that there are no memory parity errors.